

Manu S Rao

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EXPERIENCE

Software Development Engineer I

Aug 2024 – Present

Hyperface Technologies

Bangalore

- **Designed a scalable, event-driven nudge system** processing **1M+ notifications** across SMS, Email, and WhatsApp by unifying **multiple event sources and data schemas** into a single pipeline.
- **Developed a rule engine** to evaluate if incoming events qualify for a nudge, and implemented **Cooldown and Do-Not-Disturb (DND) logic** to manage message delivery while maintaining **high system efficiency**.
- **Designed and delivered a scalable, issuer agnostic backend architecture** for **Journeys PWA** by driving **core architectural decisions**, defining the **implementation strategy** and executing **end to end backend development**.
- Engineered an **optimized file ingestion pipeline** capable of processing **2M+ user records** using **asynchronous processing** and **multi-batch insertion**, reducing processing time by **75%** (from 3 hours to 45 minutes).
- **Architected a highly configurable file-ingestion workflow** supporting flexible **file pick locations**, **encryption protocols**, and **dynamic CSV column mapping**, **automating secure data transfers** and saving the Customer Success team **90 minutes per setup** while accelerating **campaign go live**.
- **Developed an automated log-file generation system** featuring **user-defined column selection**, output formatting, and a **custom query generator** supporting both tabular and fixed-variable fields; managed **scheduling and encryption** protocols, significantly **increasing bandwidth** for Project and Customer Success teams.
- Engineered an **A/B Testing (AB Config)** framework utilizing **deterministic hashing** and **percentage based bucket segmentation** to assign users to experimental variants; integrated this logic across both **automated and manual user ingestion pipelines**.

SDE Intern

Jan 2024 – Aug 2024

Hyperface Technologies

Bangalore

- **Architected a modular Android SDK** to streamline the integration of **Hyperface Web Applications and APIs**; utilized a **flavor-based architecture** to deliver optimized builds for both **Native Android** and **React Native** environments.
- **Designed and implemented a secure SIM-binding mechanism** featuring **real-time verification** via cryptographic hashes and **SMS-provider webhooks**; engineered proactive **SIM-swap detection** during every SDK initialization to mitigate account takeover and fraud risks.
- **Developed an instrumentation utility** leveraging **WebView JavaScript injection** to capture lifecycle events and API performance; implemented a **secure synchronization layer** to transmit metrics to the backend using **session token identification**.

PROJECTS

nano | *Spring Boot, MySQL*

- Developed a **high-performance URL shortener** in **Spring Boot** and **MySQL**, optimized to handle massive traffic using **sequential database indexing** and **time-based partitions**. Secured the routing system with **custom mathematical scrambling** to prevent data scraping, and integrated an in-memory **MaxMind geographic database** to track real-time user locations across **IPv4 and IPv6** networks.

TECHNICAL SKILLS

Languages: Java, Javascript

Backend Frameworks: Spring, Spring Boot, Node.js, Express.js

Frontend Frameworks: React.js, React Native, Android

Technologies: Kafka, Redis, Git, AWS (S3, Lambda, EC2)

Databases: MySQL, Amazon Redshift, Postgress

Concepts: Database Design, System Design, REST API Design

EDUCATION

RV College of Engineering

July 2024

Bachelor of Engineering in Information Science (CGPA: 9.27 / 10.00)

Bangalore, Karnataka